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Performance Report (Week 16) in IT 112: Web Systems and Technologies
2nd Semester A.Y 2025 - 2026

NEXUSWrites
“Your Tech, Your Story.”

Group Name:
Powerpuffgals

Group Members:
Kris Ann B. Apoon
April Grace R. Dacanay
Ivy Angel R. Hidalgo
Princes Mae L. Mateo
Milan R. Rellora

Course & Block:
BSIT-2A



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Introduction

The Final Project - Phase 7: Testing, Documentation, and Final Deployment represents the final stage of the group's full-stack web development project. This phase focuses on refining the system through functional testing. UI/UX improvements, debugging, documentation, and live deployment. The project integrates frontend, backend, and database technologies to create responsive and fully functional web applications capable of performing CRUD operations and handling user interactions efficiently.

Throughout this phase, the group collaborated to ensure that the system met the required standards for functionality, usability, and presentation. Various testing methods were conducted to identify and resolve errors, while GitHub and Render were used for version control and deployment.

Objectives

By the end of this phase, we were able to:

- To develop and finalize a fully functional full-stack web application.
- To ensure that all systems features, CRUD operations, and database interactions work properly through testing and debugging.
- To improve the system's user interface and user experience using responsive and modern design.
- To deploy the completed project to a live server and prepare complete project documentation.
- To enhance teamwork, collaboration, and problem solving skills throughout the project deployment process.



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I. Features Implemented

- The system includes several functional features that improve usability and overall user experience.
- **Age-Gated Secure Authentication**
Implements a secure two-step registration and login system with age verification (18+), encrypted passwords using Bcrypt.js, JWT-based session management, and custom Bootstrap error modals for improved security and user experience.

NEXUSWrites
Join the Technical Hub

ACCOUNT IDENTITY

FULL NAME
kris

EMAIL ADDRESS

Sorry! You must be at least 18 years old to join NEXUSWrites.

OK

PASSWORD
...

CONFIRM PASSWORD
...

NEXT: SETUP PROFILE >

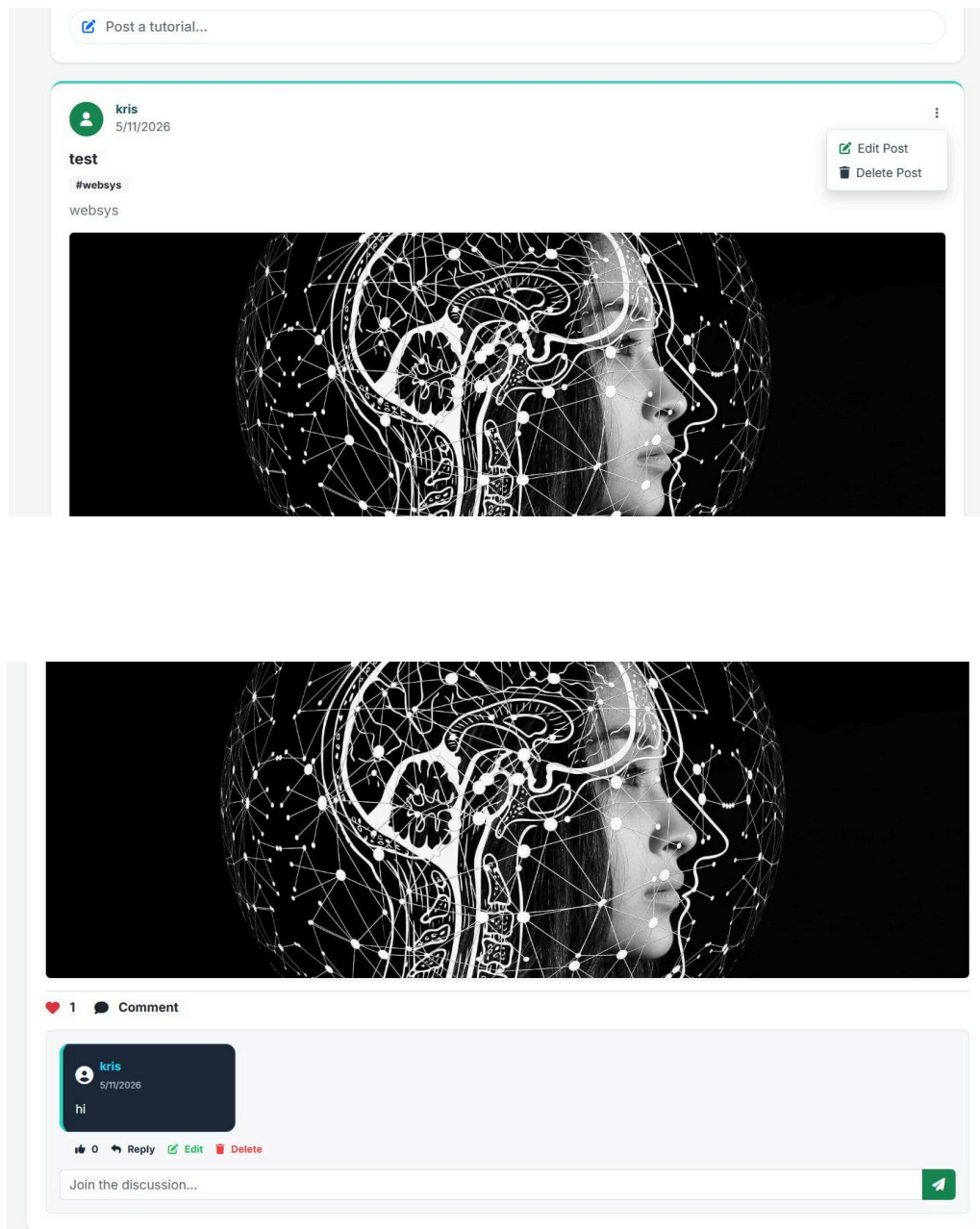


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- **Tech-Centric CRUD with Cloud Media**

Allows users to create, edit, and delete technical tutorials with categorized content, tags, and optional Cloudinary image uploads, while supporting likes and feed pagination for smoother interaction.





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Create New Tutorial

Tutorial Title

Enter an engaging title...

Category

General

Content

Write your technical story here...

Tags

e.g. JavaScript, AI, Design

Add Image

Choose File

No file chosen

Publish

Create New Tutorial

Tutorial Title

test

Category

Cybersecurity

Content

websys

Tags

websys

Add Image

Choose File

668527039_1...248306_n.jpg

Publish

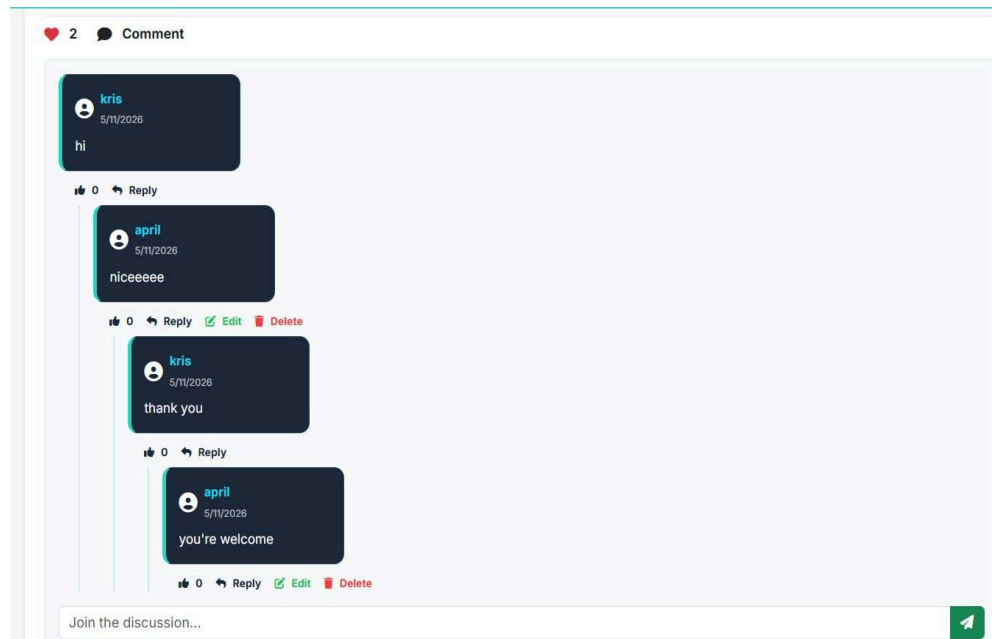


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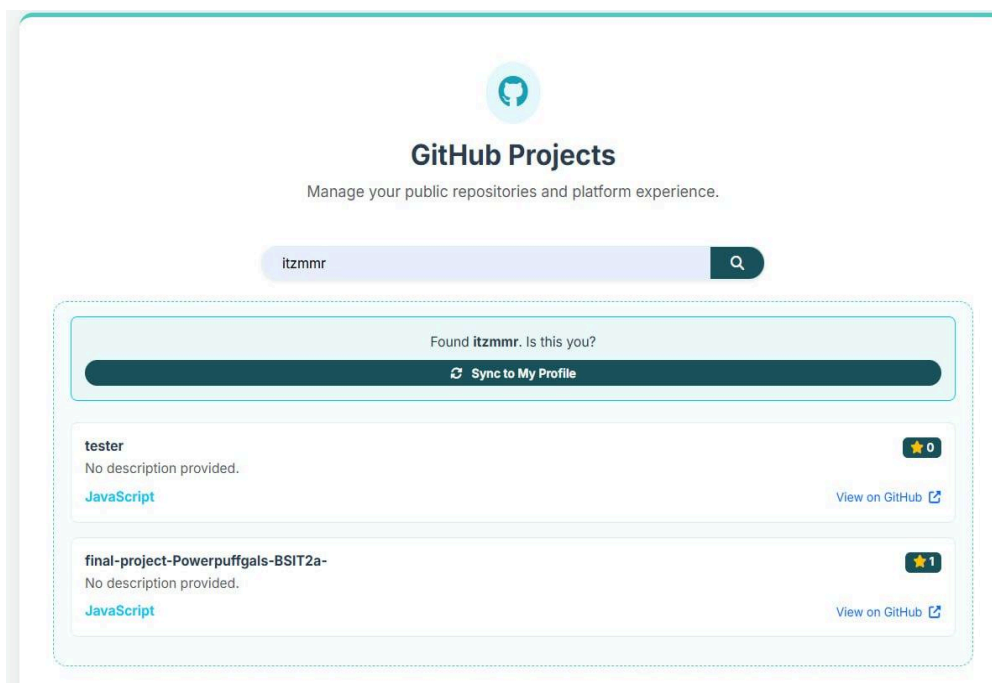
- **Hierarchical Discussion System (Nested Comments)**

Provides an interactive threaded discussion feature where users can post replies within replies, enabling unlimited nested conversations and real-time notifications for interactions.



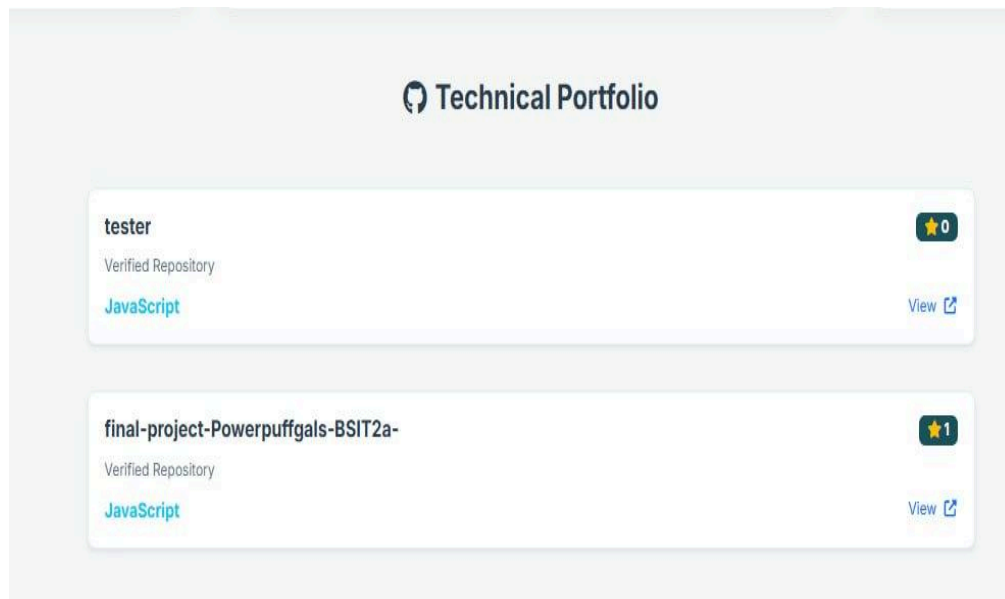
- **Live GitHub Portfolio Integration**

Integrates GitHub profiles by fetching and displaying a user's latest public repositories, allowing developers to showcase their technical portfolio directly on their profile page.



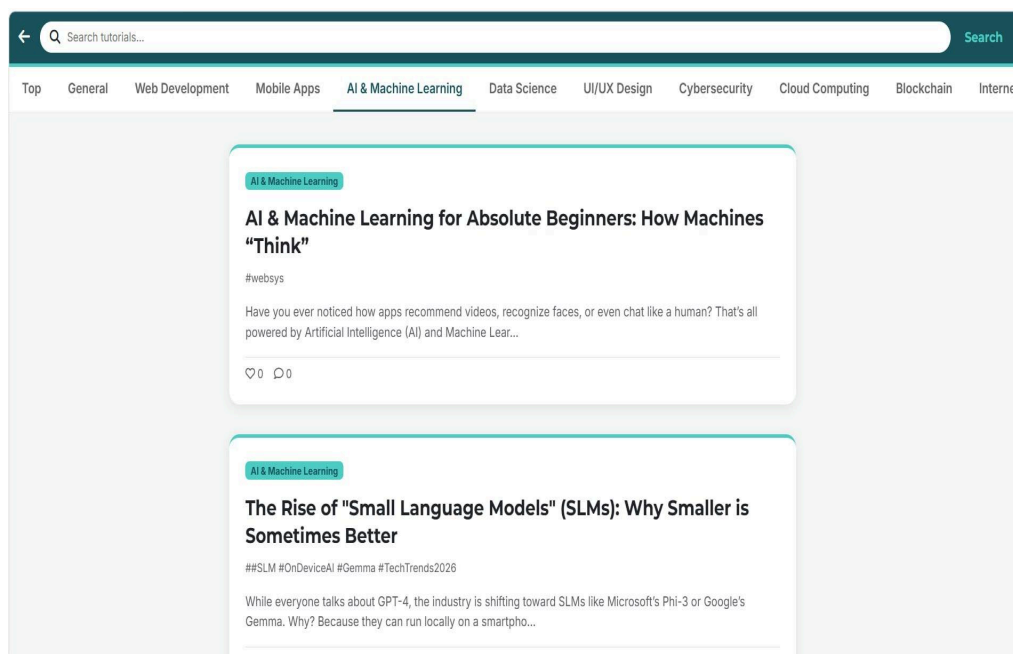


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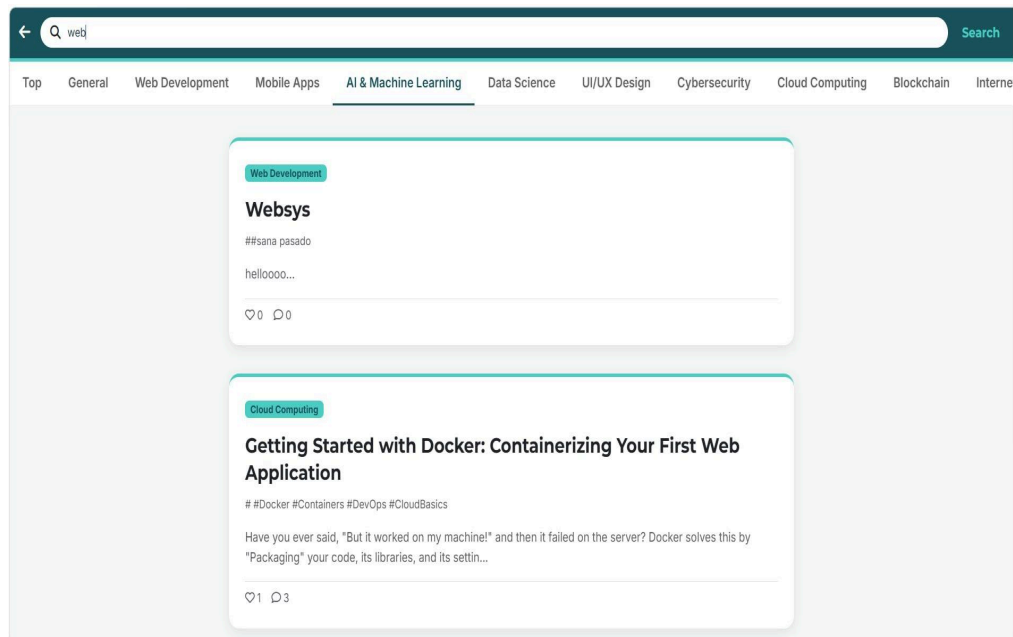
- **Dynamic Technical Categorization & Search**

Offers real-time search suggestions and category-based filtering, helping users quickly discover tutorials and content related to specific technology fields.



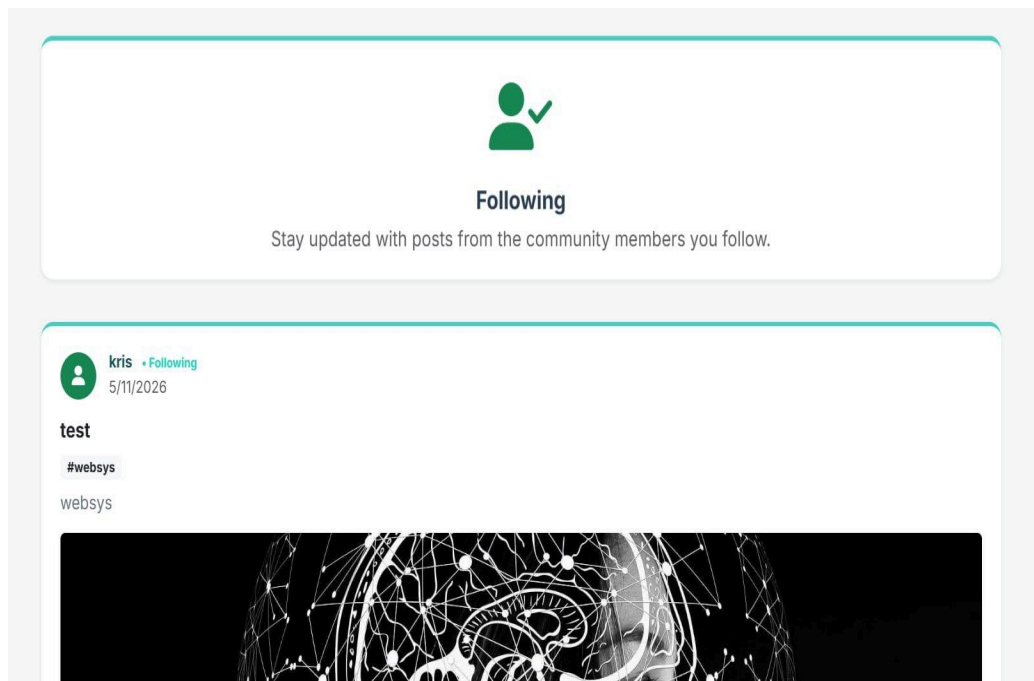


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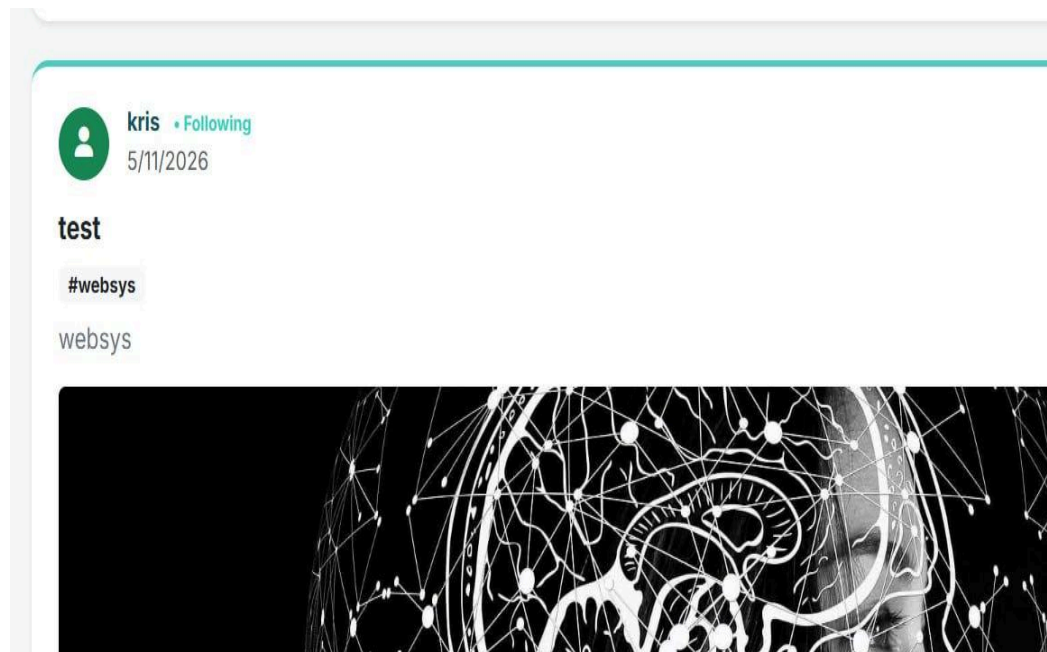
- **Following System & Personalized Feed**

-Enables users to follow or unfollow other members and access a personalized feed containing posts only from the creators they follow.



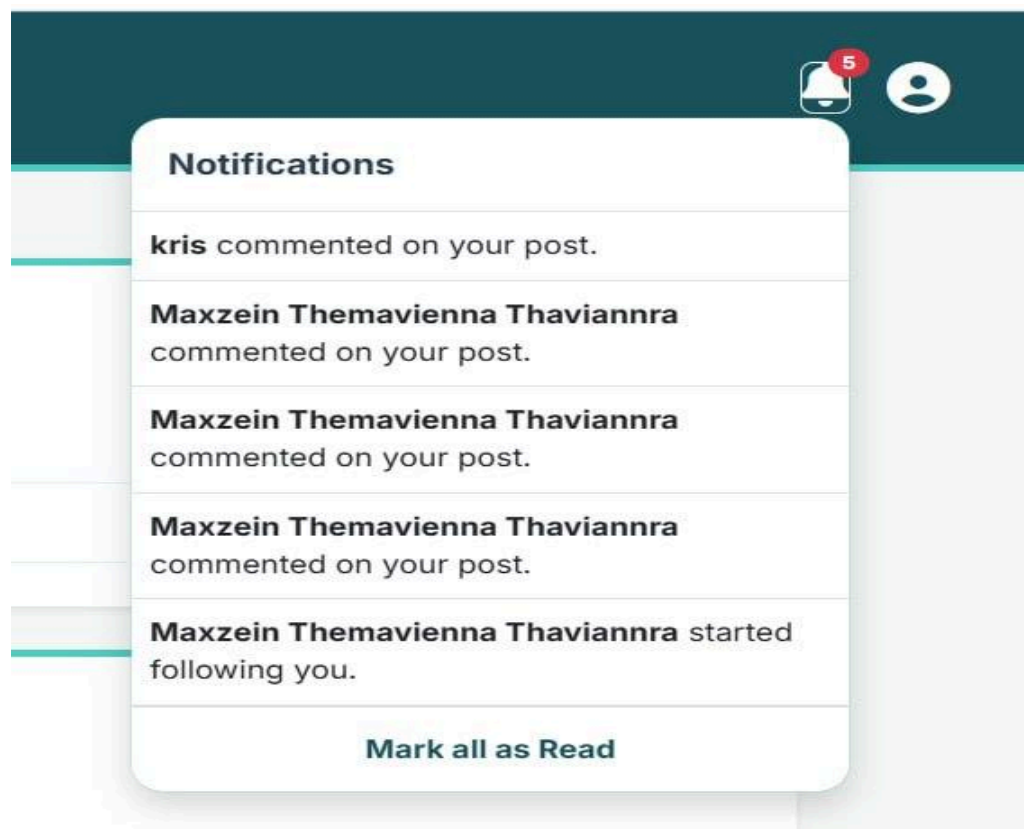


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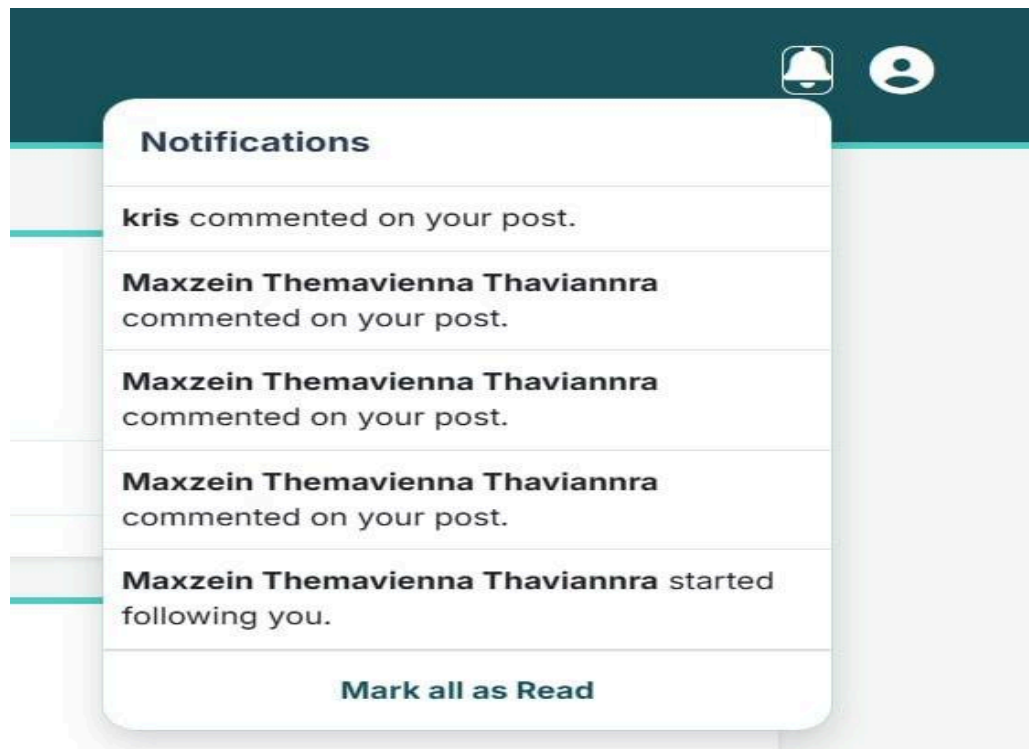
- **Real-Time Notifications**

Keeps users updated through instant notifications for likes, comments, replies, and follows, with unread badge indicators and quick read management.





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- **Settings & Personalization**

Allows users to customize their experience through adjustable themes, font sizes, language, and timezone settings, with preferences stored and applied across the platform.

1. The Update Profile modal allows developers to change their profile photo, sync their technical portfolio using a GitHub username, select a professional role, write a short bio, and check off their specific tech stack interests.



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Update Profile



Change Profile Photo

NAME *

april

EMAIL ADDRESS *

kris@gmail.com

GITHUB USERNAME

itzmmd

Enter your username to sync your technical repositories.

PROFESSIONAL ROLE

Front-End Engineer

SHORT BIO

QEEF23

INTERESTS / TECH STACK

☐ Web Systems

☐ AI & ML

☐ Cybersecurity

☒ Cloud Tech

☐ Data Analytics

☐ DevOps

☐ IoT Systems

☐ Blockchain

☐ Game Dev

☐ Open Source

Other interest? (e.g. Robotics)

Cancel

Save Changes

2.The Account Settings dashboard serves as the platform's primary navigation hub, directing users to edit their profiles, adjust display preferences, configure privacy settings, or visit the help center.

Account Settings

Manage your profile and platform experience.

 **Edit Profile**
Update your name, bio, and avatar.

 **Edit Preferences**
Customize content and display.

 **Privacy & Security**
Manage password and account visibility.

 **Help Center**
Get support and learn how to use NEXUSWrites.

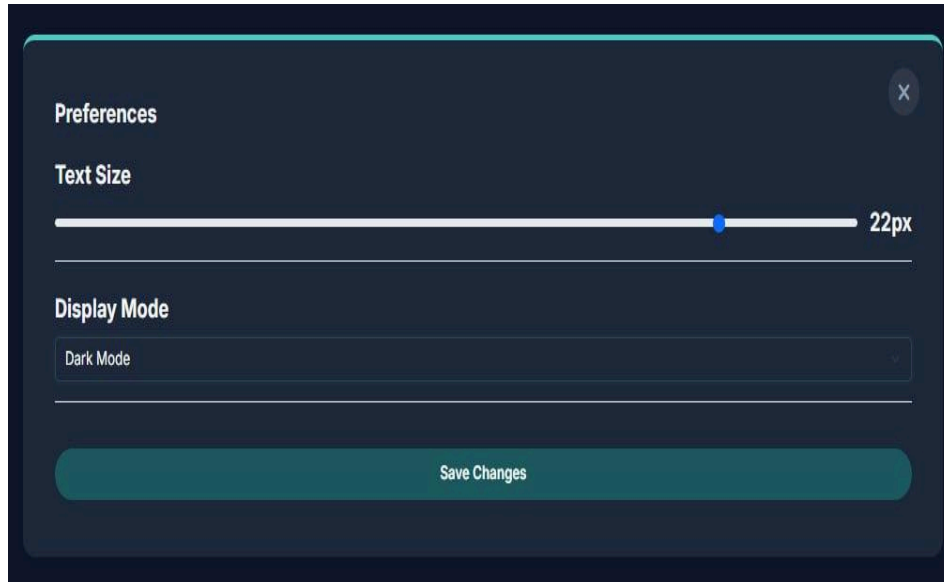
Return to Feed



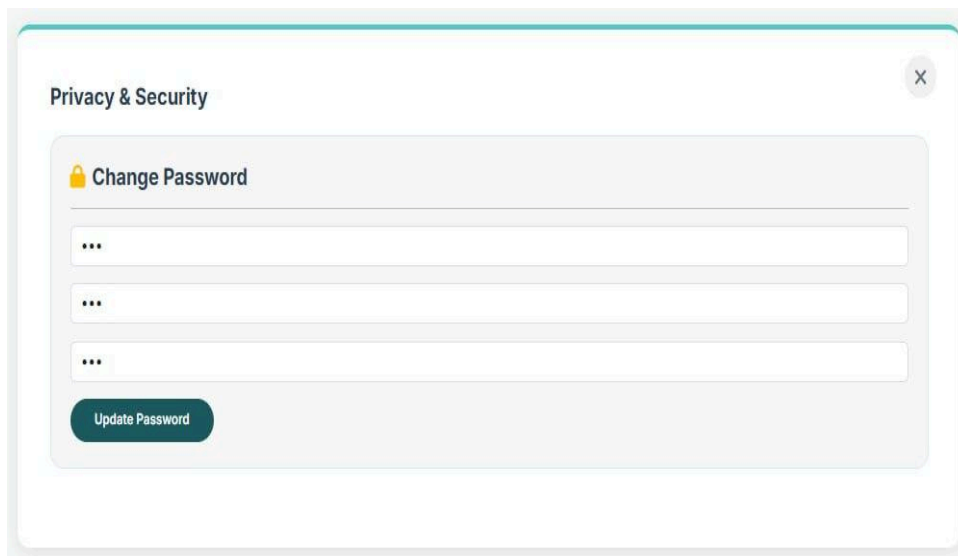
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3. The Edit Preferences panel in Dark Mode allows users to visually scale text sizes up to 22px using a slider and toggle global theme modes saved directly to their account



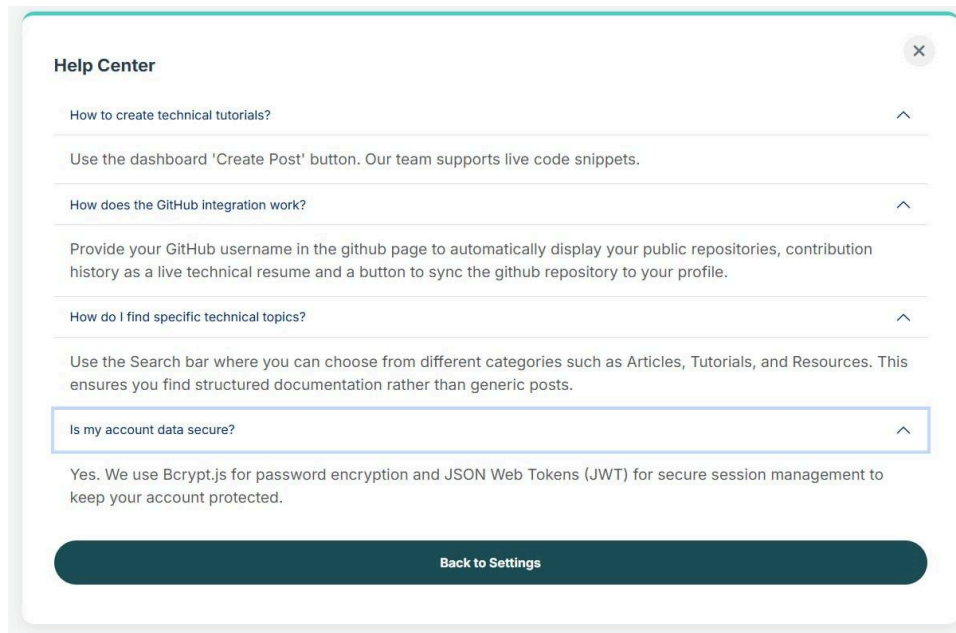
4. The Privacy & Security panel provides users with a clean, dedicated form containing three secured input fields to safely update their account password.



5. The Help Center modal features an interactive, accordion-style FAQ section that provides answers regarding technical tutorial creation, GitHub portfolio integration, topic search, and data security standards.

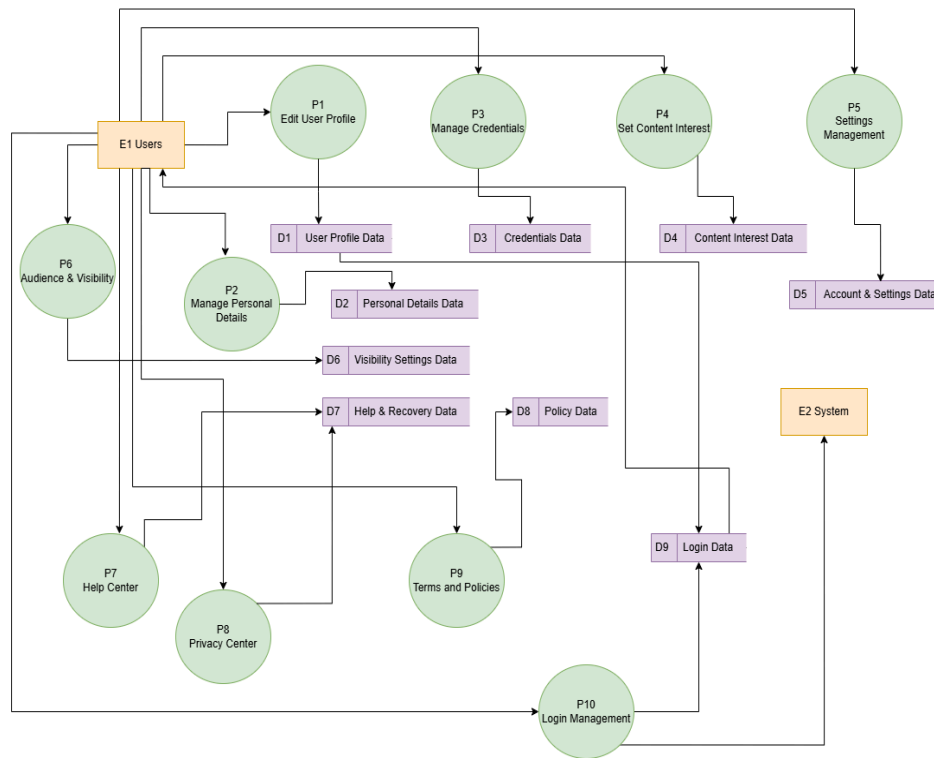


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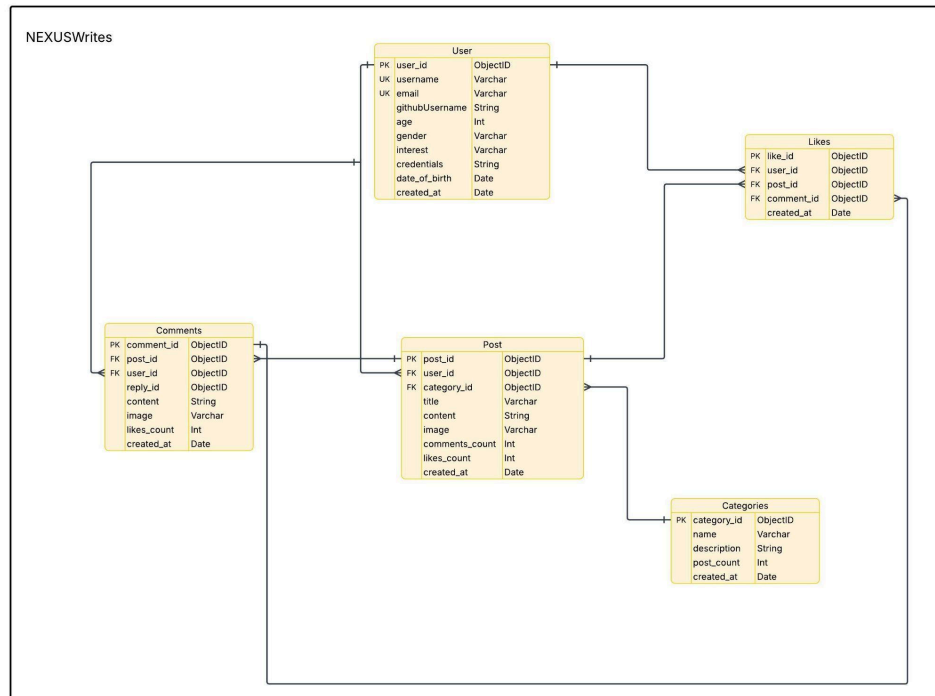
II. **Architecture Diagram**

The **Data Flow Diagram (DFD)** illustrates how data moves within the **NEXUSWrites** system. It shows the interaction between users, system processes, and data storage components to ensure efficient handling of information across the platform. The diagram includes processes such as user profile management, credential management, content interest settings, audience visibility, privacy controls, help center, policy management, and login management. It also presents how data is transferred and stored in different repositories, including profile data, credentials data, visibility settings, policy data, login data, and account settings data. This data flow helps maintain organized system operations, secure data handling, and smooth communication between users and the system.



III. Entity Relationship Diagram (ERD)

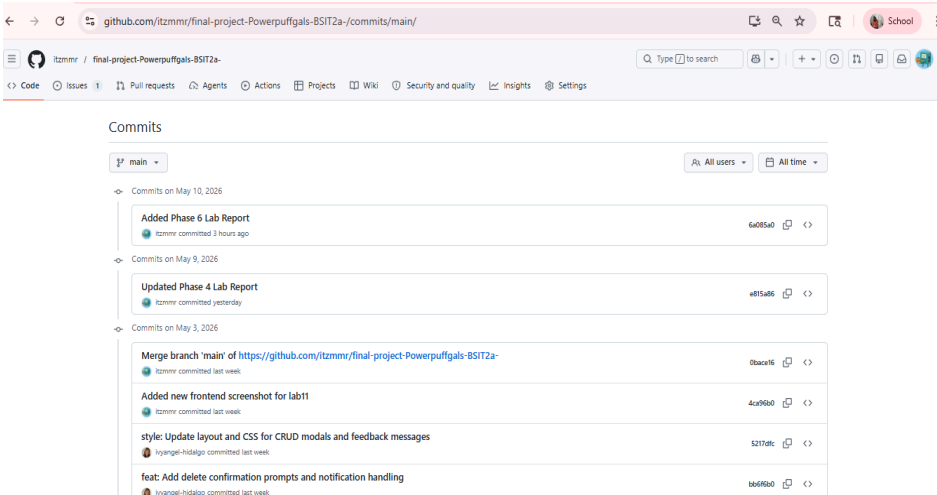
The Entity Relationship Diagram (ERD) represents the database structure of the NEXUSWrites platform and shows the relationships between the main entities of the system. The diagram includes entities such as Users, Posts, Comments, Categories, and Likes, which are connected through primary and foreign keys to maintain data integrity and organization. The Users entity stores account and profile information, while the Posts entity contains blog content created by users. The Comments entity manages user interactions within posts, and the Likes entity records engagement activities. The Categories entity helps organize blog posts into specific topics or classifications. The ERD serves as the foundation for database design, ensuring efficient data management, consistency, and scalability of the platform.



IV. GitHub Link & Commit Logs Summary

Contains the complete source code, version history, and collaboration records of the project. Final branches were merged into the main branch, unused files were removed, and documentation was finalized for deployment readiness. Commit logs demonstrate the continuous development, testing, debugging, and feature implementation performed by each team member throughout the project cycle.

<https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a-.git>



Commits

- Commits on May 10, 2026
 - Added Phase 6 Lab Report (6a085a0)
- Commits on May 9, 2026
 - Updated Phase 4 Lab Report (e85a85)
- Commits on May 3, 2026
 - Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a- (0ba0a16)
 - Added new frontend screenshot for lab11 (4a090a0)
 - style: Update layout and CSS for CRUD modals and feedback messages (5217dc)
 - feat: Add delete confirmation prompts and notification handling (1a06f0a)



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feat: Add delete confirmation prompts and notification handling hyangel-hidalgo committed last week	b6d6b0	<>
feat: Implement prefilled forms and PUT request handling for updates hyangel-hidalgo committed last week	1383ab4	<>
feat: Add edit/delete buttons and modal structures to core pages hyangel-hidalgo committed last week	23756d8	<>
feat: Implement PUT and DELETE endpoints for users, posts, and comments hyangel-hidalgo committed last week	cd87b68	<>
Added the frontend screenshots to the lab 11 folder itzmmr committed last week	5ef7335	<>
Commits on May 2, 2026		
Move to Lab Reports Folder itzmmr committed last week	8f2d580	<>
Commits on Apr 30, 2026		
Move the Phase 5 lab report to the lab reports folder. itzmmr committed 2 weeks ago	ew0252f	<>
Added Phase 5 Laboratory Report hyangel-hidalgo committed 2 weeks ago	825a044	<>
Commits on Apr 29, 2026		
style: Refined styles for dynamic content rendering hyangel-hidalgo committed 2 weeks ago	e5960d	<>
feat: Update HTML templates for dashboard and user interfaces hyangel-hidalgo committed 2 weeks ago	a7e0503	<>
feat: Implement dynamic data fetching and DOM manipulation hyangel-hidalgo committed 2 weeks ago		
1352b4b	<>	
feat: Updated API endpoints for data retrieval hyangel-hidalgo committed 2 weeks ago		
3e78771	<>	
Commits on Apr 27, 2026		
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a- hyangel-hidalgo committed 2 weeks ago	b43d9ba	<>
Implemented all frontend logic and assets for NEXUSWrites hyangel-hidalgo committed 2 weeks ago	8ee3dc6	<>
feat: Updated the notificationRoutes.js hyangel-hidalgo committed 2 weeks ago	facc4c3	<>
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a- itzmmr committed 2 weeks ago	93d89fd	<>
Added Phase 4 LabReport/Documentations itzmmr committed 2 weeks ago	fcbaw08	<>
Finalized authentication forms and brand assets for NEXUSWrites krizcan-themum committed 2 weeks ago	50d444c	<>
Sync GitHub repositories and persist user profile data in localStorage krizcan-themum committed 2 weeks ago	5339d6f	<>
Fixed translation loading order and eliminated UI initialization bugs. krizcan-themum committed 2 weeks ago	84ae5e3	<>
Updated notificationRoutes.js hyangel-hidalgo committed 2 weeks ago	e5db819	<>
Finalized server.js with complete middleware and route integration hyangel-hidalgo committed 2 weeks ago		
f9a32bc	<>	
Configured security middleware, error handling, and cloud storage hyangel-hidalgo committed 2 weeks ago		
8059382	<>	
Defined MongoDB schemas for structured content and user management hyangel-hidalgo committed 2 weeks ago		
1862e95	<>	
Establish API routing for social interactions and repository fetching hyangel-hidalgo committed 2 weeks ago		
8b4c6f1	<>	
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a- hyangel-hidalgo committed 2 weeks ago		
49627b0	<>	
Implemented all backend controllers for user auth, posts, and GitHub sync hyangel-hidalgo committed 2 weeks ago		
e8bdcf5	<>	
Added Phase 3 LabReport/Documentations itzmmr committed 2 weeks ago		
18022fe	<>	
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a- itzmmr committed 2 weeks ago		
02d40eb	<>	
Added README.md for Phase 3 Frontend itzmmr committed 2 weeks ago		
e0ba0d4	<>	
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Commits

main	All users	All time
Commits on Apr 23, 2026		
Delete frontend/pages directory	6b0c39	<>
Updated Frontend CSS Files	058d83	<>
Updated Frontend JavaScript Files	90c415	<>
Updated Frontend HTML Files	64eb4d3	<>
Added Initial Frontend styles.css file	283a4f	<>
Added initial Frontend HTML Files	4a01a9	<>
Added Initial Frontend JavaScript File	313786	<>
Commits on Apr 22, 2026		
Added Phase 2 LabReport/Documentation	95af39	<>
Commits on Apr 13, 2026		
Commits on Apr 22, 2026		
Added Phase 2 LabReport/Documentation	95af39	<>
Commits on Apr 13, 2026		
Move lab report into backend documents folder	2ba9d7b	<>
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a	742a12	<>
Backend Documentation	4745c9a	<>
Commits on Apr 12, 2026		
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a	0a7b996	<>
added README.md with routes, models, and connection setup	d2865af	<>
Mapped RESTful API endpoints to backend controllers	c48af4f	<>
Implemented backend controllers for core application logic	c38832f	<>
Developed Mongoose schemas for Users, Posts, and Interactions	68ccf9f	<>
Implemented authentication and error-handling middleware	552f8cc	<>
Commits on Mar 29, 2026		
Implemented authentication and error-handling middleware	552f8cc	<>
Configured database connection and cloud media integration	a531fec	<>
Initialized backend environment and configured .gitignore	980ca23	<>
Commits on Mar 29, 2026		
Merge branch 'main' of https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a	c1e4623	<>
Added Diagrams and Proposal for Final Project	17c111b	<>
Revise README with updated tools and project details	94f9193	<>
Add README for NEXUSWrites project overview	3d787c3	<>
Added base folders	d27d644	<>

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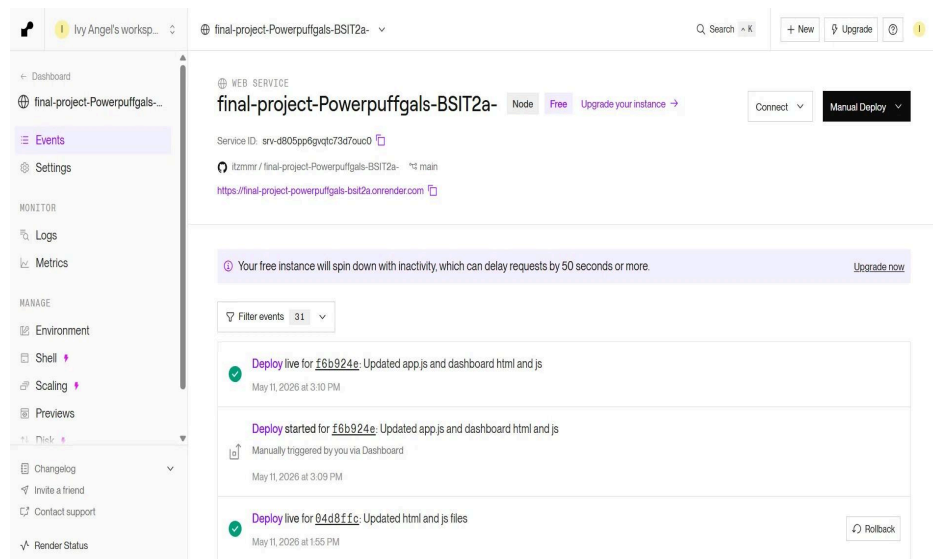


V. Render Link

The project was successfully deployed using Render to make the application publicly accessible online. The Render link and deployment screenshots serve as proof of successful deployment and live hosting

Here's the Render Link:

<https://final-project-powerpuffgals-bsit2a.onrender.com/>



VI. Summary of Division of Tasks

Throughout the project development, each member was assigned specific responsibilities to ensure the successful completion of the system. The Project Manager handled overall coordination, monitored milestones, and ensured the team stayed aligned with project goals. The Frontend Developer focused on designing and improving the user interface using HTML, CSS, and Bootstrap, while also handling responsiveness, accessibility, and UI functionality. The Backend Developer managed the server-side logic, API routes, CRUD operations, and ensured all endpoints functioned properly.

The Database Manager was responsible for creating and maintaining the MongoDB database structure, ensuring schema consistency, and verifying accurate data updates. The GitHub Manager handled repository management, branch merging, commit organization, and final repository cleanup including the README preparation. Meanwhile, the Documentation Officer prepared diagrams, screenshots, testing reports, and technical documentation for every phase of development.



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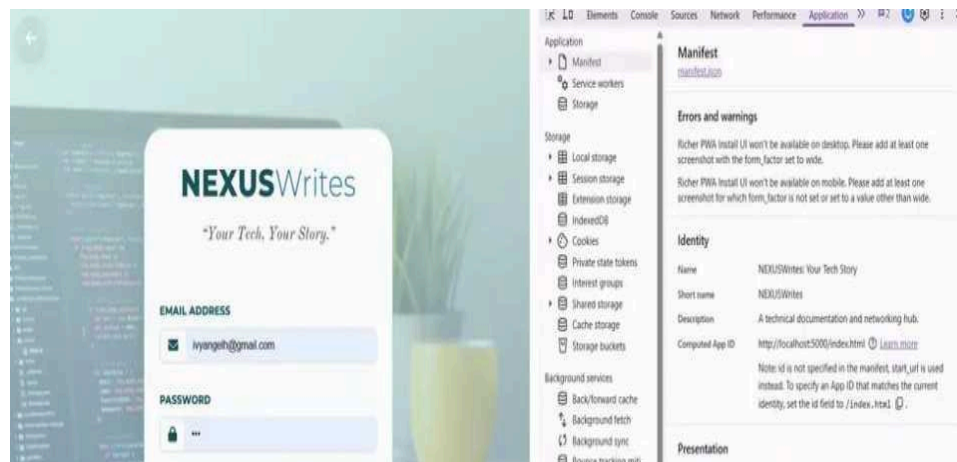


As the project progressed through Phases 1 to 7, the team continuously collaborated in testing, debugging, deployment, and refinement to ensure the system was fully functional and ready for the final project defense.

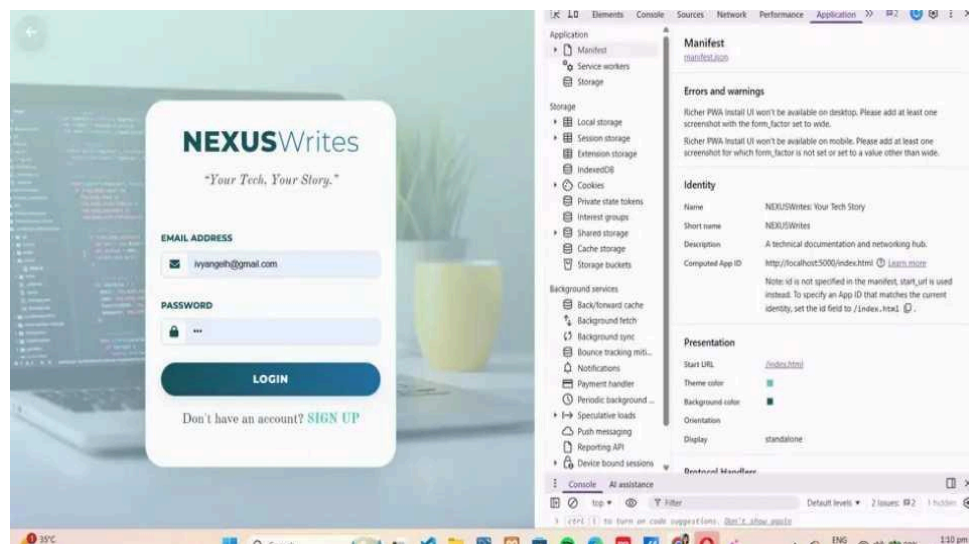
VII. Bug Testing Results/Final Test Notes

Functional and usability testing were conducted to identify and resolve system issues before final deployment. The group tested all GET, POST, PUT, and DELETE routes, validated form submissions, checked responsiveness across devices, and verified database consistency.

This screenshot displays the landing page of NEXUSWrites alongside the browser's developer console in the "Application" tab, illustrating an early debugging stage where no web app manifest has been detected yet.



This screenshot shows the secure sign-in interface next to the active Chrome DevTools, demonstrating a successfully integrated and validated manifest.json configuration that registers the platform's identity, presentation mode, and PWA capabilities.

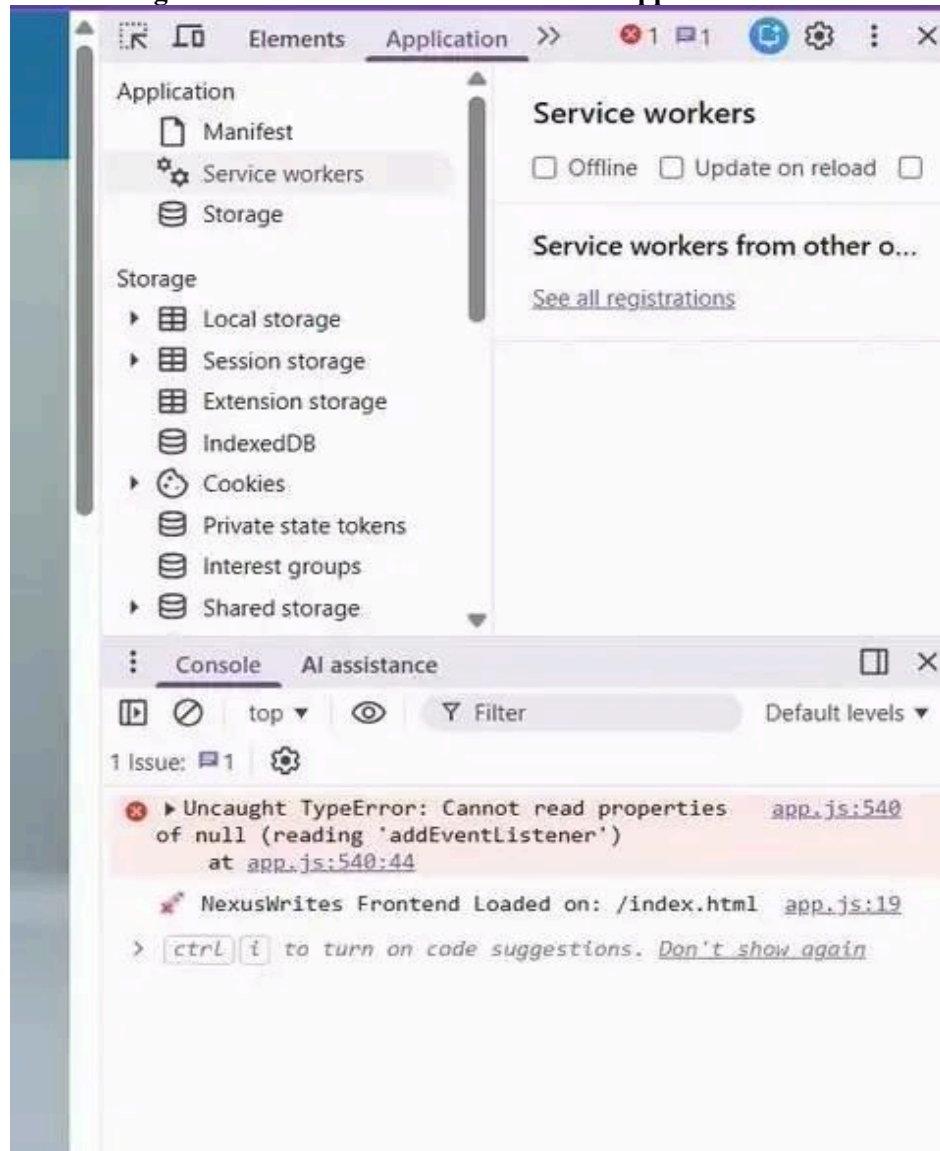




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The browser's Developer Tools console displays a JavaScript runtime error: Uncaught TypeError: Cannot read properties of null (reading 'addEventListener') located in app.js. This indicates that the system attempted to attach an event listener to an HTML element that was not yet loaded or did not exist in the DOM, causing the service worker registration process to fail during the initial execution of the frontend application.

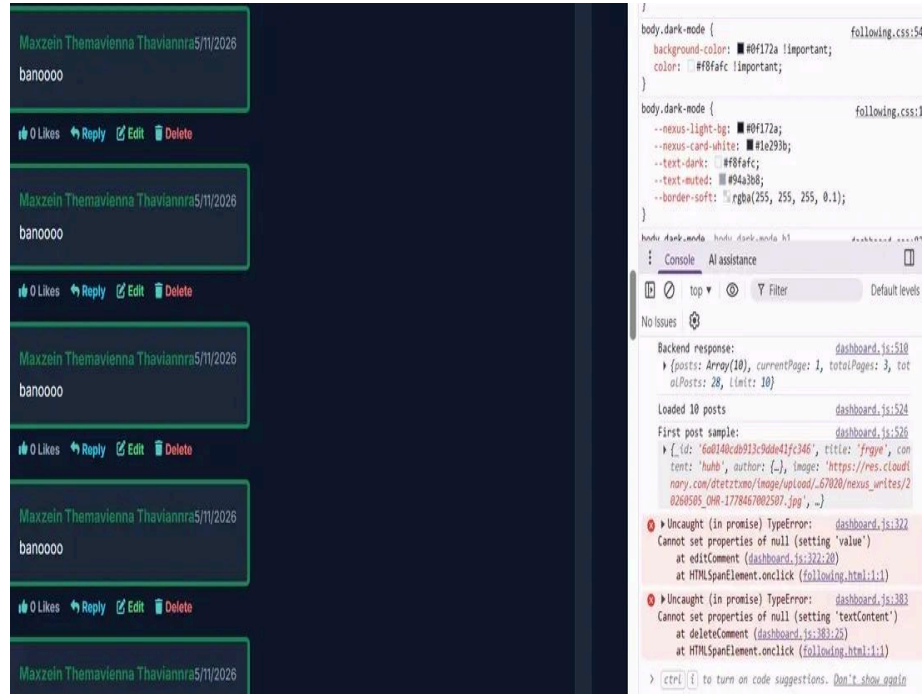




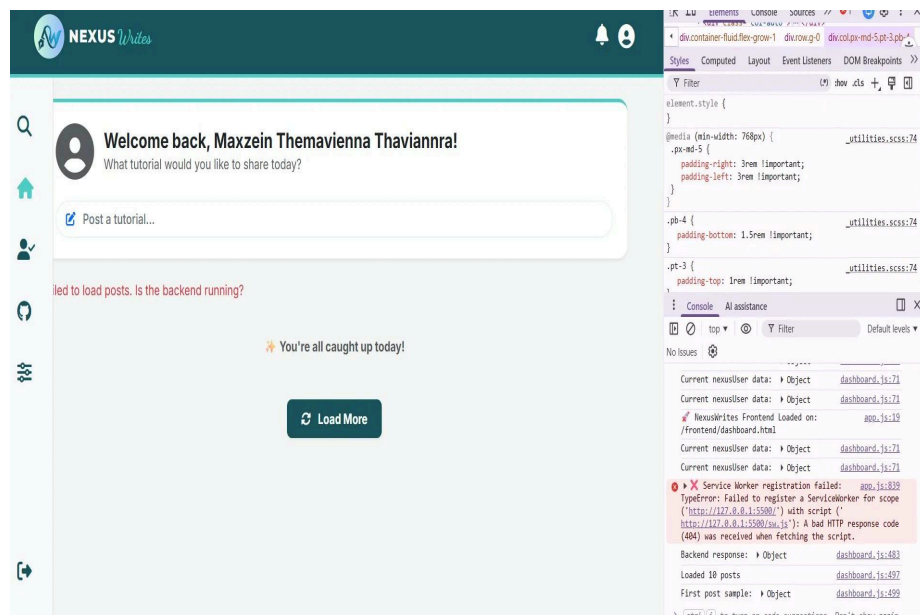
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The edit and delete modal was not functioning properly, preventing users from updating or removing comments when the buttons were clicked.



The dashboard failed to load posts because the frontend could not connect properly to the backend service worker (sw.js), resulting in a 404 error and preventing post data from displaying correctly.



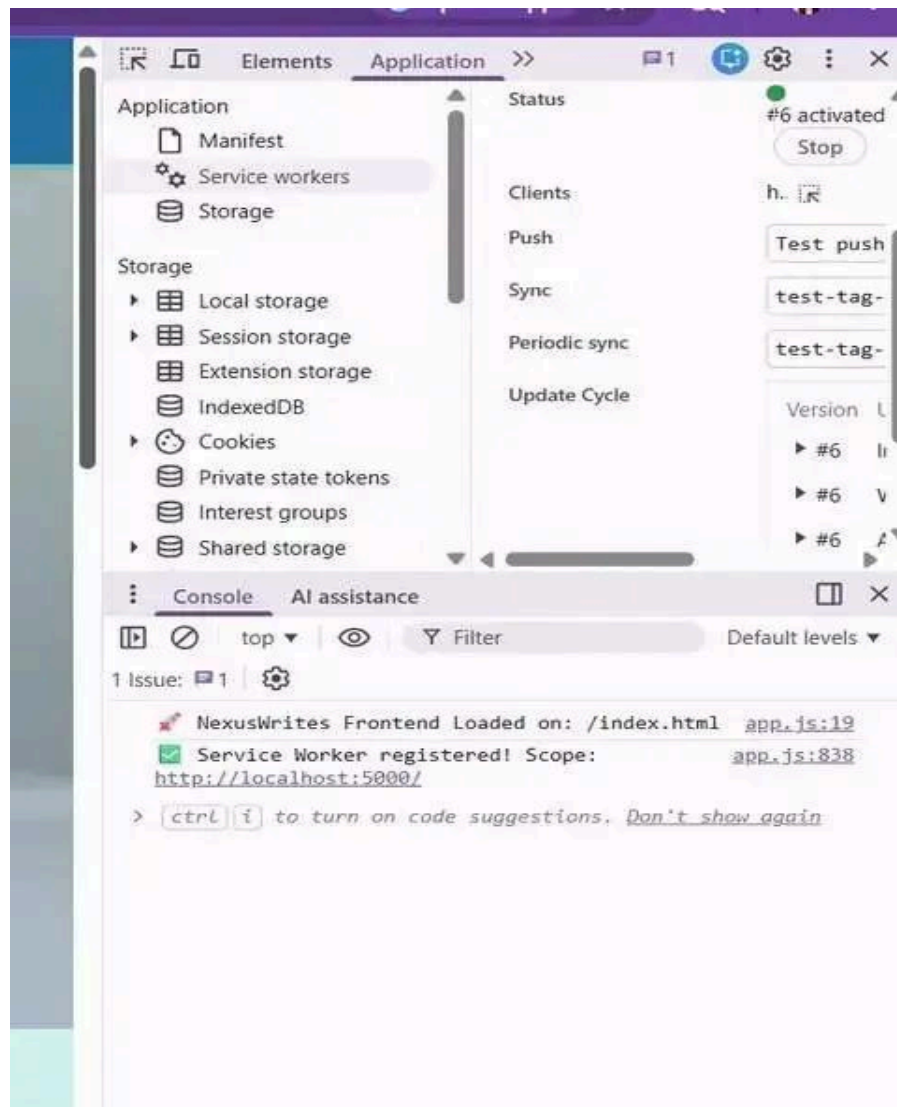


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RESULTS:

After debugging and correcting the JavaScript issue, the console confirms that the service worker was successfully registered with the message Service Worker registered! Scope: <http://localhost:5000/>. The Application tab now shows the service worker in an activated state, verifying that the Progressive Web Application (PWA) functionality is operational and capable of handling caching, offline support, and background processes correctly.

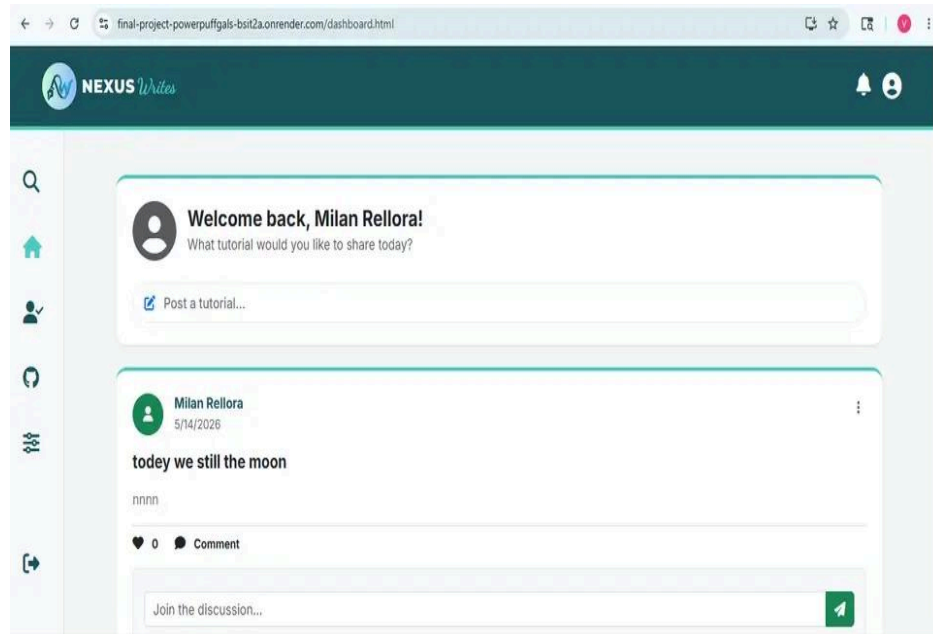




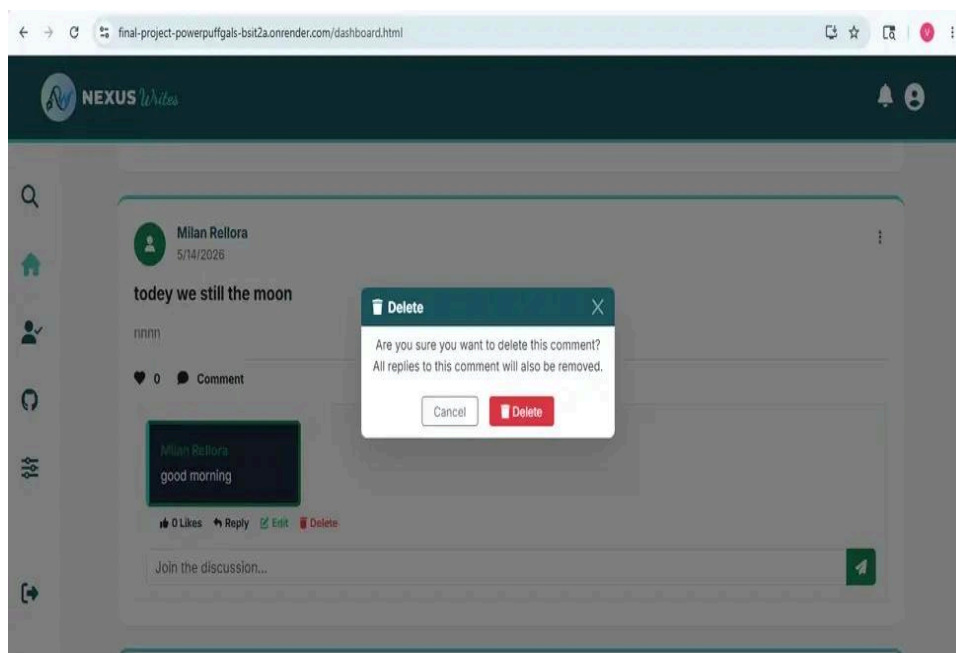
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The homepage loaded successfully and displayed all content correctly after the issue was fixed.

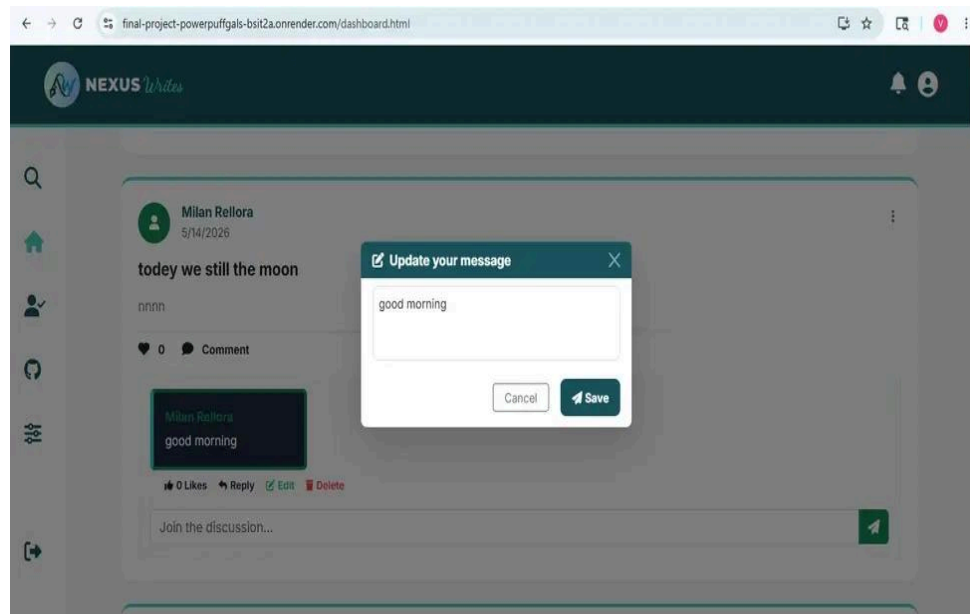


Comments can now be edited and deleted successfully through the fully functional modal feature.





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VIII. Member Reflections

Ivy Angel Hidalgo (Project Manager & Backend Developer)

As the project lead and backend developer, I was responsible for coordinating the team while also ensuring that all system endpoints worked properly. I managed the overall progress of the project, tracked remaining tasks, and prepared the system for the final defense. At the same time, I tested and confirmed that all CRUD operations and routes (GET, POST, PUT, DELETE) functioned correctly and handled invalid inputs properly. This role helped me improve both my leadership skills and my understanding of backend system logic and reliability.

Kris Ann Apoon (Frontend Developer)

As the frontend developer, I focused on polishing the UI by improving layout, responsiveness, and fixing visual bugs. I made sure all components like forms, modals, and navigation worked smoothly across devices. This experience helped me better understand user experience and interface consistency.

April Grace Dacanay (Documentation Officer)

As the documentation officer, I prepared the final reports, diagrams, and testing documentation needed for the defense. I ensured all project processes were clearly written and easy to understand. This helped me improve my ability to organize technical information clearly and professionally.



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Princes Mae Mateo (Database Manager)

As the database manager, I checked the consistency of all data and ensured the schema matched the system requirements. I verified that all updates, inserts, and deletions were properly reflected in the database. This helped me become more careful and accurate in handling structured data.

Milan Rellora (GitHub Manager)

As the GitHub manager, I merged final branches, cleaned up the repository, and ensured the README was complete and organized. I also handled the final release tag and made sure the project structure was ready for submission. This role improved my organization and version control management skills.